The admissible edge sets of the next layer form a matroid that does not depend on the chosen maximally sparse set

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Editorial note. The surplus-independence lemma at source lines 676-726 is the article's own source-local support result and is delivered with its complete proof; it is already prepared in the corpus as problem 142659 and is not a newly counted target here.

Editorial note. The two counting lemmas invoked inside that support proof are supplied by their statements only (source lines 600-611 and 631-640), exactly as in the prepared item 142659; their proofs belong to that other unit and are not part of this task.

Editorial note. The context span at source lines 564-578 begins with the notation sentence that defines V_i(F); it continues a preceding line of the article that also points forward to the matroid theorem, which is not part of this task.

Editorial note. No retained line contains a citation, a figure or an included file, so the package declares neither a bibliography nor an asset.

Editorial note. The retained environments print with this package's own numbering rather than the numbering of the article's full document.

Editorial scope. One original exact result: the matroid well-definedness of the admissible edge family of a layer, with the article's own complete surplus-independence lemma as source-local support.

Allowed standard prerequisites (editorial).

- Matroid independence axioms. A hereditary family with the augmentation property, equivalently a family determined by its circuits; no rank formula is assumed.
- Maximally sparse sets and the surplus. Maximally sparse edge sets on V_1 union ... union V_j and the minimum-defining surplus functional theta_{alpha_j} are supplied by the retained context spans (source lines 399-406, 564-578); membership in I_j is the criterion displayed at 575-577, not the naive edge count.

Definition 1. *Let $G = (V, E)$ be a (multi)graph such that V has a partition into subsets $V_1, V_2, \ldots, V_n$, where every edge $e \in E$ has endpoints in V_i and V_{i+1} for some $i \in \{1, \ldots, n-1\}$.*

Let $K = (k_1, \ldots, k_n) \in \mathbb{R}^n_{>0}$ *and* $L = (l_{1,2}, \ldots, l_{n-1,n}) \in \mathbb{R}^{n-1}_{\geq 0}$. *For any* $V' \subseteq V$, *we define*

$$l(V') = \sup_{\substack{i: V_i \cap V' \neq \emptyset \\ V_{i+1} \cap V' \neq \emptyset}} \{l_{i,i+1}\}.$$

We say that G *is* (K, L)*-sparse if, for any subset* $V' \subseteq V$, *one has*

$$|E(V')| \leq \sum_{i=1}^{n} k_i |V' \cap V_i| - l(V').$$

We note that the use of sup *makes* $l(V') = -\infty$ *whenever the defining set is empty.*
Furthermore, G *is* (K, L)*-tight if* G *is* (K, L)*-sparse, and additionally*

$$|E| = \sum_{i=1}^{n} k_i |V \cap V_i| - \sup_{i=1,\ldots,n-1} \{l_{i,i+1}\}.$$

We furthermore say that a subset of edges $E' \subseteq E$ *is* (K, L)*-sparse or* (K, L)*-tight if the graph* (V, E') *is* (K, L)*-sparse or* (K, L)*-tight, respectively. We say that a subset of edges* $E' \subseteq E$ *is maximally sparse if* E' *is* (K, L)*-sparse, and if for any edge* $vw \in E \setminus E'$ *such that* $v, w \in V(E')$, $E' \cup \{vw\}$ *is not* (K, L)*-sparse.*

We will mainly consider the case where the entries of K and L are integers, and we need the following inequalities to hold for the (K, L)-sparse graphs to be a family of graphs that is well-behaved.

$$k_i + k_{i+1} > l_{i,i+1}, \tag{1}$$
$$k_{i+1} + l_{i-1,i} > l_{i,i+1}, \tag{2}$$
$$l_{1,2} \leq \ldots \leq l_{n-1,n}. \tag{3}$$

As stated in the introduction, we will show that one does, however, get a greedoid. We will consider strings $e_1 \ldots e_m$ on the alphabet E. For any string $\alpha = e_1 \ldots e_m$, we define the edge set $E_\alpha := \{e_1, \ldots, e_m\}$. Then, we define the collection $\mathcal{L}_{(K,L)}$ to be the set of strings on E that satisfy the following three properties:

1. For any $\alpha = e_1 \ldots e_m$, the set E_α is (K, L)-sparse.
2. For any $\alpha = e_1 \ldots e_m$ and $1 \leq i < j \leq n-1$, if $e_{k_1} \in E(V_i \cup V_{i+1})$ and $e_{k_2} \in E(V_j \cup V_{j+1})$, then $k_1 < k_2$.
3. For any $\alpha = e_1 \ldots e_m$ and $i \in \{1, \ldots, n-1\}$ such that there is an e_j with $e_j \in E(V_i \cup V_{i+1})$, the subgraph $E_\alpha \cap E(V_1 \cup \cdots \cup V_i)$ is maximally (K, L)-sparse on $V_1 \cup \cdots \cup V_i$.

We will use the terminology *maximal feasible set* to mean a set E_α, where $\alpha \in \mathcal{L}_{(K,L)}$ is a maximal word, meaning there is no word $\beta \in \mathcal{L}_{(K,L)}$ such that $\beta = \alpha e$ for some e. Expanding on what this means, the maximal feasible sets E' are the sets $E' \subseteq E$ such that $E'(V_1 \cup \cdots \cup V_j)$ is a maximally sparse set for each $j \in \{2, \ldots, n\}$.

To prove that $\mathcal{L}_{(K,L)}$ is a greedoid, we will show that the sets of edges that can be added to a word α between V_j and V_{j+1} are the independent sets of a

matroid. Let us make this precise. Suppose that E^* is a maximally sparse set supported on $V_1 \cup ... \cup V_j$, and let $\mathcal{I}_j$ be the collection of subsets $E' \subseteq E(V_j, V_{j+1})$ such that $E^* \cup E'$ is (K, L)-sparse. We will show that the elements of $\mathcal{I}_j$ are the independent sets of a matroid with ground set $E(V_j, V_{j+1})$. We will moreover show (see Lemma 4) that $\mathcal{I}_j$ does not depend on the chosen maximally sparse set. We denote the pair $(E(V_j, V_{j+1}), \mathcal{I}_j)$ by $\mathcal{M}_j$.

for any subset of edges $F \subseteq E$, we define $V_i(F) := V(F) \cap V_i$. Let α_j be a word in $\mathcal{L}_{(K,L)}$ such that no $e \in E(V_{j+1} \cup V_{j+2})$ appears in α_j, and E_{α_j} is maximally sparse on $V_1 \cup \cdots \cup V_{j+1}$. We then define, for subsets $X \subseteq V_{j+1}$ the *surplus*

$$\theta_{\alpha_j}(X) := \min_{W : X \subseteq W} \{(\sum_{i=1}^{j+1} k_i |V_i \cap W| - |E_{\alpha_j}(W)|\}.$$

We remark that θ depends on the chosen parameters K.We will sometimes suppress the subscript for α_j, writing α in stead of α_j. It follows easily from the definition that if $X \subseteq Y$ then $\theta_\alpha(X) \leq \theta_\alpha(Y)$.

Let us briefly motivate the definition of the surplus. Fix some $\alpha \in \mathcal{L}_{(K,L)}$ such that $E_\alpha \subseteq E(V_1 \cup \cdots \cup V_j)$ is a maximally sparse subset of $E(V_1 \cup \cdots \cup V_j)$. Defining $\mathcal{I}_j$ using E_α, one can see that a set $E' \subseteq E(V_j \cup V_{j+1})$ belongs to $\mathcal{I}_j$ if and only if

$$|F| \leq k_{j+1} |V_{j+1}(F)| + \theta_\alpha(V_j(F)) - l_{j,j+1} \tag{4}$$

for any subset $F \subseteq E'$.

Lemma 2. *Let $K \in \mathbb{Z}^n_{>0}, L \in \mathbb{Z}^{n-1}_{\geq 0}$ be such that inequalities (1), (2) and (3) are valid. Let $G = (V, E)$ be a graph such that the vertex set has a partition $V = V_1 \cup \cdots \cup V_n$ such that all edges $e \in E$ have endpoints in $V_i \cup V_{i+1}$ for some $1 \leq i < n$.*

Suppose also that $B \subseteq E$ is (K, L)-sparse. Let $X \subseteq V_1 \cup \cdots \cup V_j$ and $T \subseteq V_1 \cup \cdots \cup V_j$ be such that $X \cap T$ contains some vertices of V_{j-1} and V_j. Suppose also that

$$\sum_{i=1}^{j} k_i |V_i \cap T| - |B(T)| = \ell_{j-1,j}.$$

Then

$$\sum_{i=1}^{j} k_i |V_i \cap X| - |B(X)| \geq \sum_{i=1}^{j} k_i |V_i \cap (T \cup X)| - |B(T \cup X)|.$$

Lemma 3. *Let $K \in \mathbb{Z}^n_{>0}, L \in \mathbb{Z}^{n-1}_{\geq 0}$ be such that inequalities (1), (2) and (3) are valid. Let $G = (V, E)$ be a graph such that the vertex set has a partition $V = V_1 \cup \cdots \cup V_n$ such that all edges $e \in E$ have endpoints in $V_i \cup V_{i+1}$ for some $1 \leq i < n$, and assume that G is (K, L)-sparse.*

Let $B_1, B_2 \subseteq E$ be tight edge sets, with $V(B_i) \subseteq V_1 \cup \cdots \cup V_t$. If $V(B_1) \cap V(B_2) \cap V_{t-1} \neq \emptyset$ and $V(B_1) \cap V(B_2) \cap V_t \neq \emptyset$, then $B_1 \cap B_2$ and $B_1 \cup B_2$ are tight subgraphs. Moreover, one has

$$V(B_1 \cup B_2) = V(B_1) \cup V(B_2)$$
$$V(B_1 \cap B_2) = V(B_1) \cap V(B_2)$$

Lemma 4. *Let $K \in \mathbb{Z}^n_{>0}, L \in \mathbb{Z}^{n-1}_{\geq 0}$ be such that inequalities* (1), (2) *and* (3) *are valid. Let $G = (V, E)$ be a graph such that the vertex set has a partition $V = V_1 \cup \cdots \cup V_n$ such that all edges $e \in E$ have endpoints in $V_i \cup V_{i+1}$ for some $1 \leq i < n$.*

Suppose that $\mathcal{M}_c$ is a matroid for all $c < j$. Then the set $\mathcal{I}_j$ is independent of the maximally sparse set E^*. Furthermore, $\mathcal{I}_j$ is non-empty whenever $E(V_j \cup V_{j+1})$ is non-empty.

Lemma 5. *Let $K \in \mathbb{Z}^n_{>0}, L \in \mathbb{Z}^{n-1}_{\geq 0}$ be such that inequalities* (1), (2) *and* (3) *are valid. Let $G = (V, E)$ be a graph such that the vertex set has a partition $V = V_1 \cup \cdots \cup V_n$ such that all edges $e \in E$ have endpoints in $V_i \cup V_{i+1}$ for some $1 \leq i < n$.*

If $\mathcal{M}_c$ is a matroid for all $1 \leq c \leq j$, then $\theta_{\alpha_j} = \theta_{\alpha'_j}$ for every two words α_j and α'_j, such that E_{α_j} and $E_{\alpha'_j}$ are maximal feasible subsets supported on $V_1 \cup \cdots \cup V_{j+1}$.

Proof. We are given α_j and α'_j. For any word $\alpha_j = e_1, \ldots e_m$ such that E_{α_j} is supported on $V_1 \cup \cdots \cup V_{j+1}$ and $d < j$, define α_d to be the word $e_1 \ldots e_{\tilde{m}}$ obtained by restricting to those edges supported on $V_1 \cup \cdots \cup V_{d+1}$. I.e. α_d is the word resulting from deleting the edges $e_{\tilde{m}+1} \ldots e_m$ where $\tilde{m}+1$ is the smallest index such that for all $i \geq \tilde{m}+1$, one has $e_i \in E(V_{d+i} \cup V_{d+i+1})$ for some $i \geq 1$. In particular, α_0 will always be the empty word and for subsets $X \subseteq V_1$, that $\theta_{\alpha_0}(X) = \theta_{\alpha'_0}(X) = k_1|X|$, which serves as the basis for our induction. Suppose now that the statement holds for all $d < j$. We will show it holds for $d = j$.

Let $X \subseteq V_{j+1}$. One can verify that one can express θ_{α_j} as a function of $\theta_{\alpha_{j-1}}$. Namely, one has:

$$\theta_{\alpha_j}(X) = \min_{\substack{W \subseteq V_j \cup V_{j+1} \\ X \subseteq W}} k_{j+1}|V_{j+1} \cap W| + \theta_{\alpha_{j-1}}(V_j \cap W) - |E_{\alpha_j}(W)|.$$

By induction, we can assume that $\theta_{\alpha_{j-1}} = \theta_{\alpha'_{j-1}}$. Thus, it suffices to consider the case where α_j and α'_j only differ for edges $e \in E(V_j \cup V_{j+1})$, and we may assume $\alpha_j \neq \alpha'_j$ since otherwise the lemma is trivially true. In this case, since we know that $\mathcal{M}_j$ is a matroid, we only need to show that in the case where $E_{\alpha_j} = \left(E_{\alpha'_j} \cup \{e\}\right) \setminus \{f\}$, one has $\theta_{\alpha_j}(X) = \theta_{\alpha'_j}(X)$, since the statement then follows for all maximally chosen words α_j and α'_j. Suppose that the added edge, e, has endpoints v, w. By Lemma 3, we know that the intersection

$$M = \bigcap_{\substack{T \subseteq E_{\alpha'_j}, \\ v,w \in V(T), \\ T \text{ tight}}} T$$

is a nonempty tight subset of $E_{\alpha'_j}$. Furthermore, if f were not in M, then M would be a tight subset of E_{α_j}, as f is the only edge of $E_{\alpha'_j}$ not in E_{α_j}. Then, $M \cup \{e\}$ would not be (K, L)-sparse, which contradicts E_{α_j} being (K, L)-sparse. It follows that $f \in M$. Now, let $X \subseteq V$, and suppose W minimises the expression for $\theta_{\alpha'_j}$, so

$$\theta_{\alpha'_j}(X) = \sum_{i=1}^{j+1} k_i|V_i \cap W| - |E_{\alpha'_j}(W)|.$$

If $f \notin E_{\alpha'_j}(W)$, then we see that

$$|E_{\alpha_j}(W)| \geq |E_{\alpha'_j}(W)|$$

so it follows immediately that

$$\begin{aligned}\theta_{\alpha_j}(X) &\leq \sum_{i=1}^{j+1} k_i|V_i \cap W| - |E_{\alpha_j}(W)| \\ &\leq \sum_{i=1}^{j+1} k_i|V_i \cap W| - |E_{\alpha'_j}(W)| = \theta_{\alpha'_j}(X).\end{aligned}$$

If $f \in E_{\alpha'_j}(W)$, then we get using Lemma 2, that

$$\sum_{i=1}^{j+1} k_i|V_i \cap W| - |E_{\alpha'_j}(W)| \geq \sum_{i=1}^{j+1} k_i|V_i \cap (W \cup V(M))| - |E_{\alpha'_j}(W \cup V(M))|,$$

and hence we may assume that $V(M) \subseteq W$.

Since $E_{\alpha_j} = \left(E_{\alpha'_j} \cup \{e\}\right) \setminus \{f\}$, and e and f are both supported on $V(M)$ it follows that,

$$|E_{\alpha'_j}(W \cup V(M))| = |E_{\alpha_j}(W \cup V(M))|.$$

It now follows that

$$\theta_{\alpha_j}(X) \leq \sum_{i=1}^{j+1} k_i|V_i \cap W| - |E_{\alpha'_j}(W)| = \theta_{\alpha'_j}(X).$$

We conclude that $\theta_{\alpha_j}(X) \leq \theta_{\alpha'_j}(X)$ in all cases. Switching the roles of E_{α_j} and $E_{\alpha'_j}$, the same argument gives that $\theta_{\alpha_j}(X) \geq \theta_{\alpha'_j}(X)$, so $\theta_{\alpha'_j}(X) = \theta_{\alpha_j}(X)$. □

Proof. First, assuming that $E^* \cap E(V_{j-1} \cup V_j)$ is non-empty, it is easy to see, using inequality (2), that $E^* \cup \{e\}$ is (K, L)-sparse for any edge $e \in E(V_j \cup V_{j+1})$. If $E^* \cap E(V_{j-1} \cup V_j) = \emptyset$, then it is easy to see, using inequality (1), that $E^* \cup \{e\}$ is (K, L)-sparse for any edge $e \in E(V_j \cup V_{j+1})$. Hence, $\mathcal{I}_j$ is non-empty whenever $E(V_j \cup V_{j+1})$ is non-empty.

Let E_1^* and E_2^* be two maximally sparse subsets of $E(V_1 \cup \cdots \cup V_j)$, and let $\alpha_{j-1}, \alpha'_{j-1} \in \mathcal{L}_{(K,L)}$ be such that $E_{\alpha_{j-1}} = E_1^*$, $E_{\alpha'_{j-1}} = E_2^*$. Suppose that $E \subseteq E(V_j \cup V_{j+1})$, and that $E \cup E_1^*$ is sparse. Our goal is to show that $E \cup E_2^*$ is also sparse. We make a few remarks about $E \cup E_1^*$ first. Let $F \subseteq E$ and consider $V_j(F)$ and $V_{j+1}(F)$. We note that for any subset $A \subseteq E_1^*$, one has, by sparsity of $F \cup A$:

$$|F \cup A| = |F| + |A| \leq \Sigma_{i=1}^{j-1} k_i|V_i(A)| + k_j|V_j(A \cup F)| + k_{j+1}|V_{j+1}(F)| - l_{j,j+1}.$$

Hence, one has

$$|F| \leq \Sigma_{i=1}^{j-1} k_i|V_i(A)| + k_j|V_j(A \cup F)| + k_{j+1}|V_{j+1}(F)| - l_{j,j+1} - |A|.$$

Since A is arbitrary, we see that:

$$|F| \leq k_{j+1}|V_{j+1}(F)| - l_{j,j+1} + \min_A \{\Sigma_{i=1}^{j-1} k_i |V_i(A)| + k_j |V_j(A \cup F)| - |A|\}.$$

It is then clear that:

$$|F| \leq k_{j+1}|V_{j+1}(F)| - l_{j,j+1} + \theta_{\alpha_{j-1}}(V_j(F)).$$

Now, for any subset $C \subseteq E_2^*$, we see that

$$\begin{aligned}
|F \cup C| &= |F| + |C| \\
&\leq k_{j+1}|V_{j+1}(F)| - l_{j,j+1} + \theta_{\alpha_{j-1}}(V_j(C \cup F)) + |C| \\
&= k_{j+1}|V_{j+1}(F)| - l_{j,j+1} + \theta_{\alpha'_{j-1}}(V_j(C \cup F)) + |C| \\
&\leq k_{j+1}|V_{j+1}(F)| - l_{j,j+1} + \Sigma_{i=1}^{j-1} k_i |V_i(C)| + k_j |V_j(C \cup F)| \\
&- |C| + |C| \\
&\leq \Sigma_{i=1}^{j+1} k_i |V_i(C \cup F)| - l_{j,j+1}
\end{aligned}$$

We applied the observation that if $X \subseteq Y$ then $\theta_{\alpha_{j-1}}(X) \leq \theta_{\alpha_{j-1}}(Y)$ in the second line, with $X = V_j(F)$ and $Y = V_j(C \cup F)$. We have used Lemma 5 to go from the second line to the third. Since any subset of $E_2^* \cup E$ is of the form $F \cup C$, this shows that $E_2^* \cup E$ is sparse, completing the proof of the Lemma. $\square$